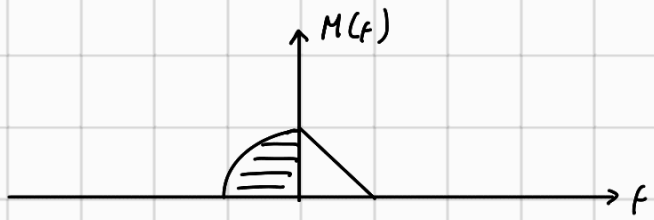
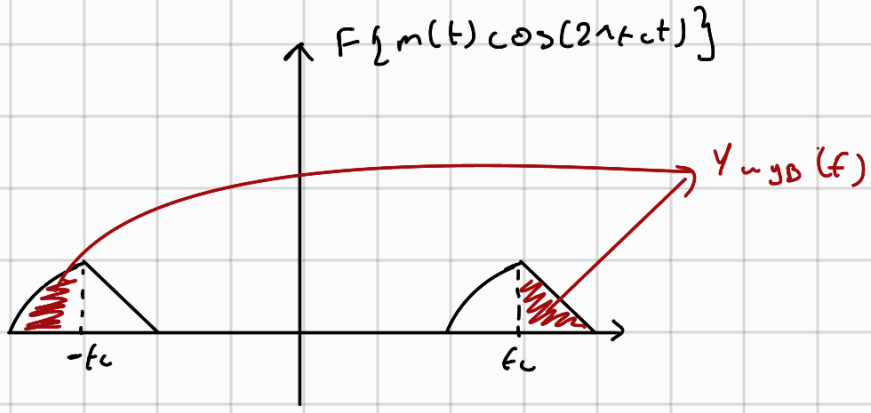
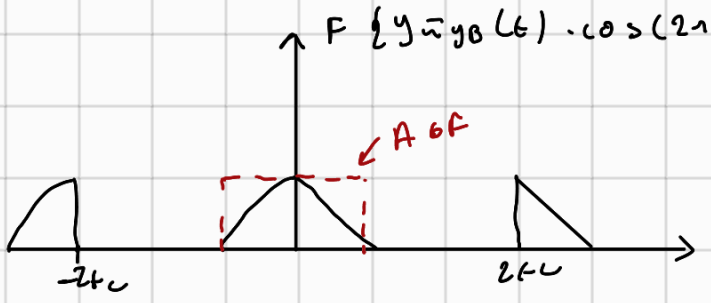
TYB - GM (SSB-Am)



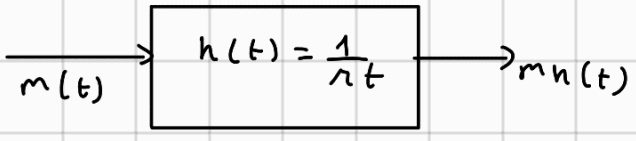




$$y_{TYB}(t) = A_c\left[m(t)\cos(2\pi f_c t) \pm m_n(t)\sin(2\pi f_c t)\right]$$

(+ → AYB, − → iiYB; $m_n(t)$ → m(t)'nin Hilbert dönüşümü (Prolcisite $\hat{m}(t)$))

$$m_n(t) = m(t) \otimes \underbrace{\frac{1}{\pi t}}_{h(t)}$$



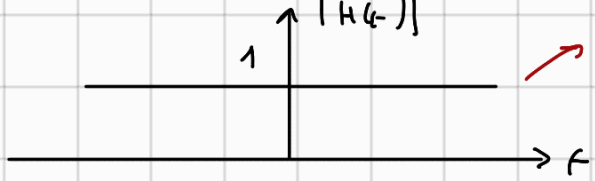
$$m_n(t) = \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{m(z)}{t-z}\,dz$$

$$h(t) = \frac{1}{\pi t} \xrightarrow{F} H(f) = -j\,Sgn(f) = \begin{cases} -j, & f>0 \\ j, & f<0 \end{cases} = \begin{cases} e^{-j\pi/2} & f>0 \\ e^{j\pi/2} & f<0 \end{cases}$$

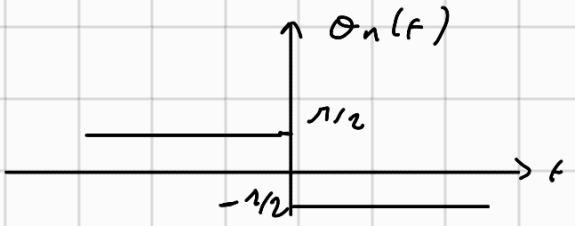
$$|H(f)| = 1$$

$$\theta_n(f) = \begin{cases} -\pi/2 & f>0 \\ \pi/2 & f<0 \end{cases}$$

⟹



tüm geçiren filtre



Bu filtre 90 derece faz farkı eklemek

$$P_{y_{TYB}} = \frac{A_c^2}{2} P_m + \frac{A_c^2}{2} P_{m_n}$$ → oda P_m'e eşit genlik değişmediği için

$$= A_c^2 P_m$$

$$y_{AYB}(t) + y_{\tilde{A}YB}(t) = 2 A_c\, m(t) \cos(2\pi f_c t) = y(t)$$

Örnek

$$m(t) = \cos(2\pi f_m t)$$

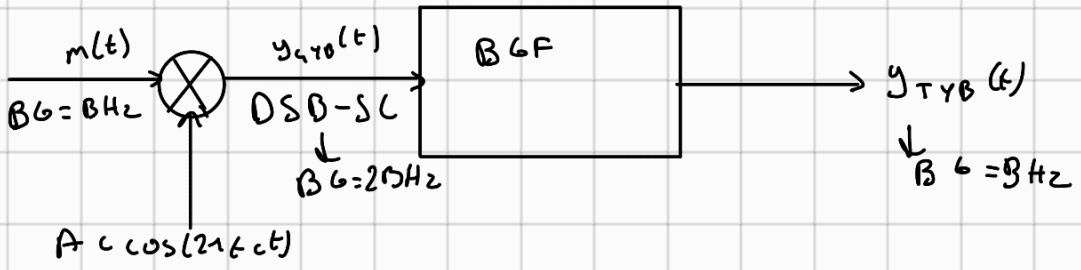
$$m_h(t) = \cos(2\pi f_m t - \pi/2) = \sin(2\pi f_m t)$$

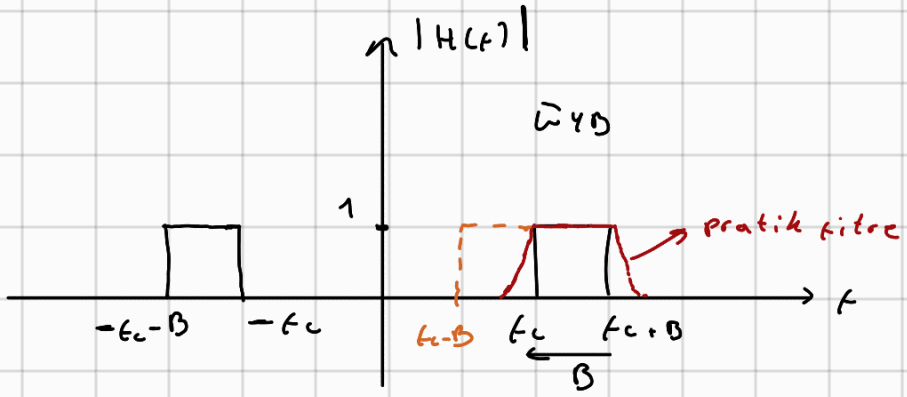
$$y_{TYB}(t) = A_c \left[\cos(2\pi f_m t)\cdot\cos(2\pi f_c t) \mp \sin(2\pi f_m t)\sin(2\pi f_c t)\right]$$

$$= A_c \cos\left[2\pi (f_c \mp f_m) t\right]$$

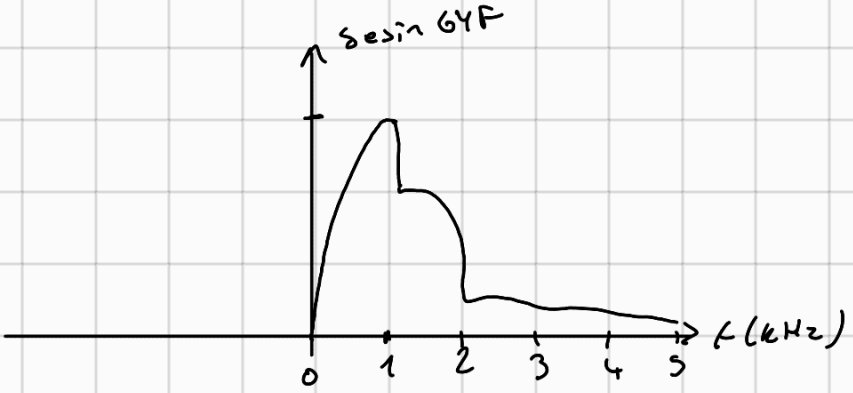
TYB - GM modülatörleri (ek örnekli)

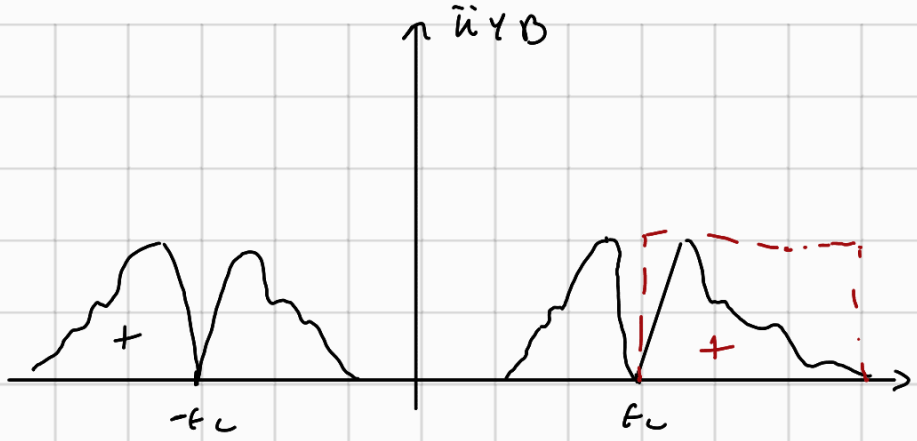
1) Filtreleme tekniği (Filtering)





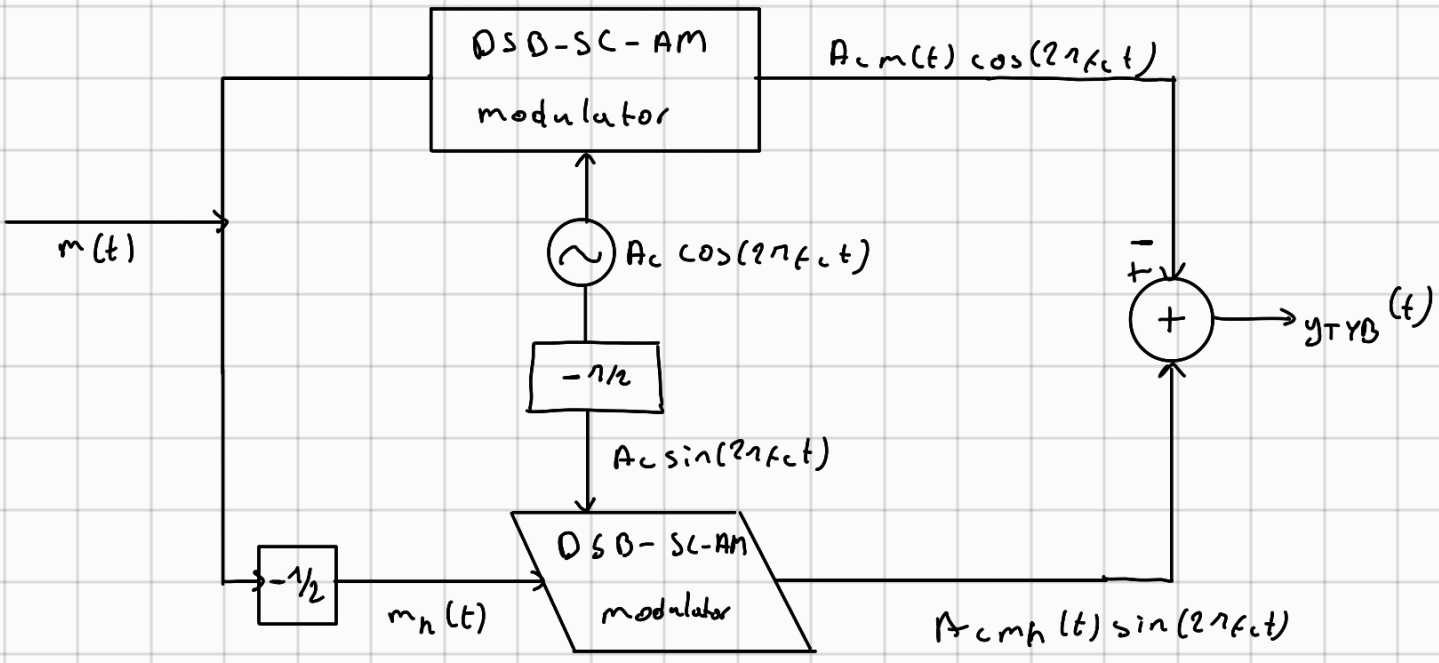
Mesaj işaretinin 0Hz civarındaki frekanslarda bileşenleri çok yakın olmalıdır





Sesin anlamlı bileşenleri 300-3500 Hz

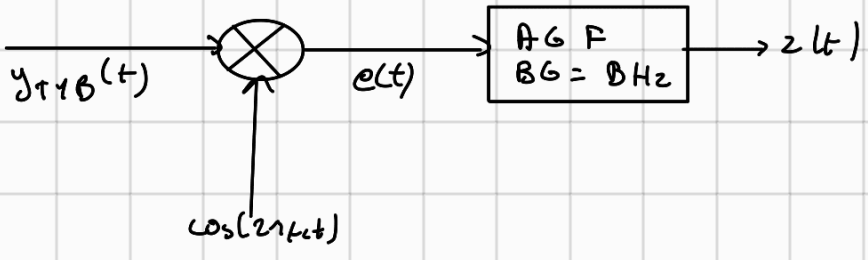
2) Faz öteleme tekniği



hilbert filtresini tüm noktalara 90 derece çizdirmek gerçekte zordur

TYB - GM - Demodülatörleri

1) Faz tahminli demodülasyon (Coherent demodulation)



$$y_{TYB}(t) = A_c m(t)\cos(2\pi f_c t) \mp A_c m_h(t)\sin(2\pi f_c t)$$

$$e(t) = y_{TYB}(t)\cdot\cos(2\pi f_c t)$$
$$= \frac{A_c}{2} m(t)\left[\underbrace{\cos(0)}_{1} + \cos(2\pi 2f_c t)\right] \mp \frac{A_c}{2} m(t)\left[\underbrace{\sin(0)}_{0} + \sin(2\pi 2f_c t)\right]$$

$$e(t) = \frac{A_c}{2} m(t) + \frac{A_c}{2}\underbrace{\left[m(t)\cos(2\pi 2f_c t) \mp m_h(t)\sin(2\pi 2f_c t)\right]}_{2f_c\text{'de TYB}}$$

↙ AGF'den geçmez

$$z(t) = \frac{A_c}{2} m(t)$$

* Sınav sorusu alıcıyı tahmin ederken θ kadar fazı farklı

$$e(t) = y_{TYB}(t)\cdot\cos(2\pi f_c t)$$
$$= \frac{A_c}{2} m(t)\left[\cos(\theta) + \cos(2\pi 2f_c t + \theta)\right] \mp \frac{A_c}{2} m(t)\left[\sin(\theta) + \sin(2\pi 2f_c t + \theta)\right]$$

$$e(t) = \frac{A_c}{2} m(t)\cos\theta \mp \frac{A_c}{2} m_h(t)\sin\theta + 2f_c\text{'de TYB}$$

↙ AGF'den geçmez

θ mesajı bozar, θ zamana bağlıdır

Faz tahminsiz demodülasyon
(Non coherent demodulation)

Envelope detection of SSB-Am with a LC = SSB + C

$$y_{SSB+C}(t) = \underbrace{A_c\cos(2\pi f_c t)}_{LC} + \underbrace{A\left[m(t)\cos(2\pi f_c t) + m_h(t)\sin(2\pi f_c t)\right]}_{SSB}$$

$$= \left[A_c + Am(t)\right]\cos(2\pi f_c t) + Am_h(t)\sin(2\pi f_c t)$$

$$= E(t)\cos(2\pi f_c t + \theta(t))$$

$a\cos x + b\sin x = c\cos(x+\theta)$
$c = \sqrt{a^2+b^2}$ $\quad \theta = -\arctan(b/a)$

$$E(t) = \left[(A_c + Am(t))^2 + (Am_n(t))^2 \right]^{1/2}$$

$$= \left[A_c^2 + A^2 m^2(t) + 2AA_c m(t) + A^2 m_n^2(t) \right]^{1/2}$$

$$= A_c \left[1 + \frac{A^2 m^2(t)}{A_c^2} + \frac{A^2 m_n^2(t)}{A_c^2} + \frac{2Am(t)}{A_c} \right]^{1/2}$$

$A_c >> A|m(t)|$
$\Rightarrow A_c >> A|m_n(t)|$

$$\Rightarrow E(t) \cong A_c \left[1 + \frac{2Am(t)}{A_c} \right]^{1/2}$$

$$\cong A_c \left[1 + \frac{Am(t)}{A_c} \right]$$

$$= A_c + Am(t)$$

★ $|x| << 1 \Rightarrow (1+x)^n \cong 1 + nx$

ÇYB - 6M
TYD - 6M

$A_c \geq |m(t)|$
$A_c >> |m(t)|$

$\Big\}$ $V_{TYB} < V_{ÇYB}$

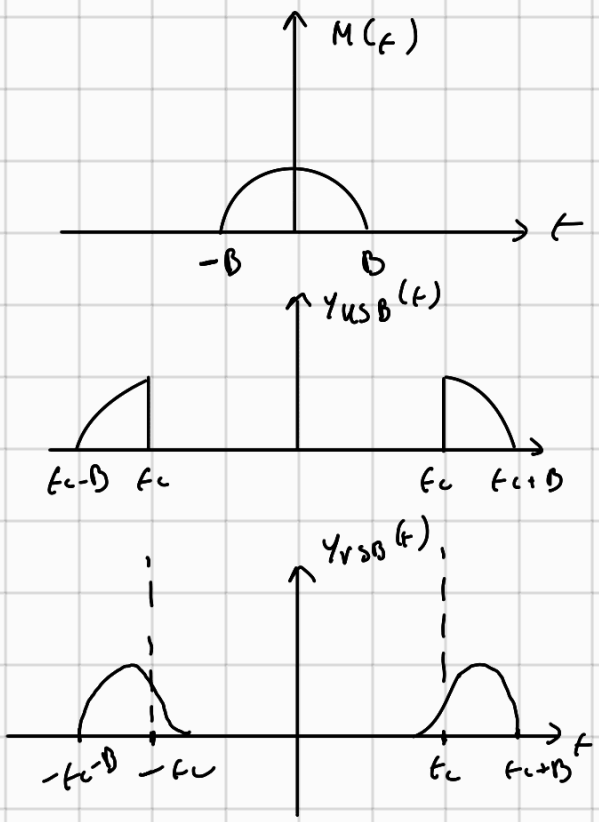
Artık Yan Band (AYB) - 6M

(Vestigial Side band (VSB) - AM)
asymmetric sideband

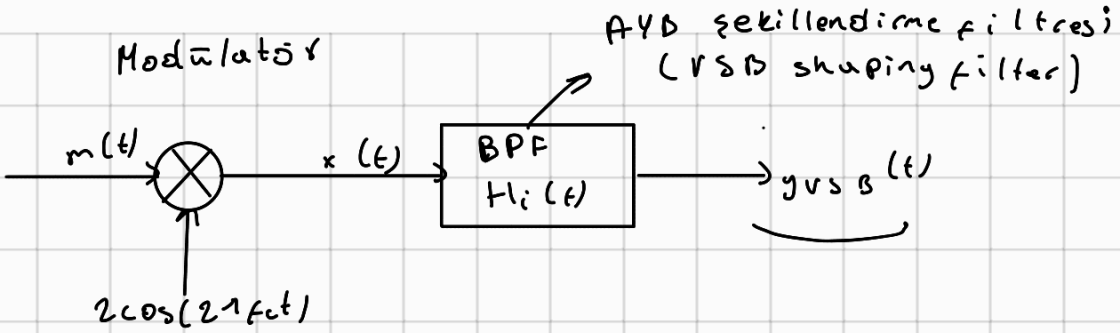
Hç band!

sıgan beyaz televizyon 6 MHz

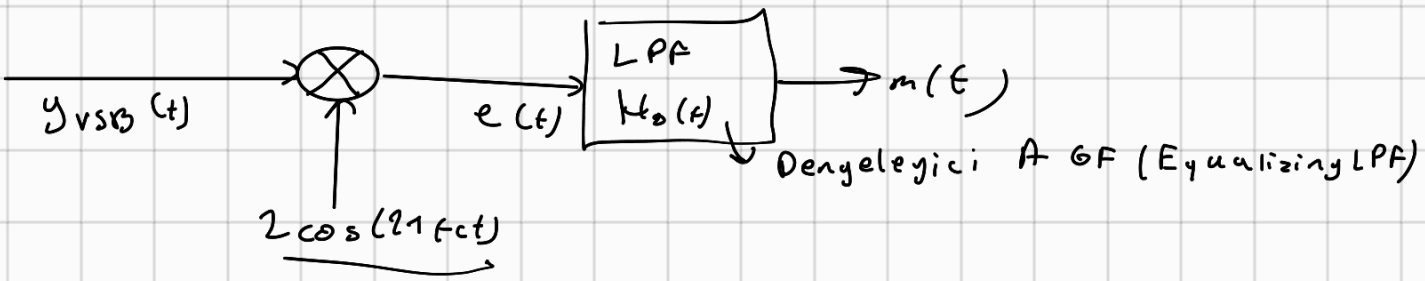
$$BG\{AYB\} = 1{,}25\ BG\{TYB\}$$



Modülatör



Demodülatör



Faz tahminli demodülasyon

$e(t) = 2 y_{VSB}(t) \cos(2\pi f_c t) \rightarrow E(f) = Y_{VSB}(f-f_c) + Y_{VSB}(f+f_c)$

$M(f) = E(f) \cdot H_o(f) = [Y_{VSB}(f-f_c) + Y_{VSB}(f+f_c)] H_o(f)$

$x(t) = 2m(t) \cos(2\pi f_c t) \rightarrow X(f) = M(f-f_c) + M(f+f_c)$

$Y_{VSB}(f) = X(f) H_i(f)$

$= M(f-f_c) H_i(f) + M(f+f_c) H_i(f)$

$M(f) = [M(f-2f_c) H_i(f-f_c) + M(f) H_i(f-f_c) + M(f) H_i(f+f_c) + M(f+2f_c) H_i(f+f_c)] H_o(f)$

AGF'den geçmezler

$M(f) = M(f) [H_i(f-f_c) + H_i(f+f_c)] H_o(f)$

$$H_o(f) = \frac{1}{H_i(f-f_c) + H_i(f+f_c)}, \quad |f| \le B \text{ iken}$$

$-B \le f \le B$

Eğer $H_i(f-f_c) + H_i(f+f_c) = 1, \quad |f| \le B$

$H_0(f) = 1, \quad |f| \leq B = \Pi\left(\frac{f}{2B}\right)$ ideal AGF



VSB+C

A_c, DSB-LC < A_c, VSB < A_c, SSB+C

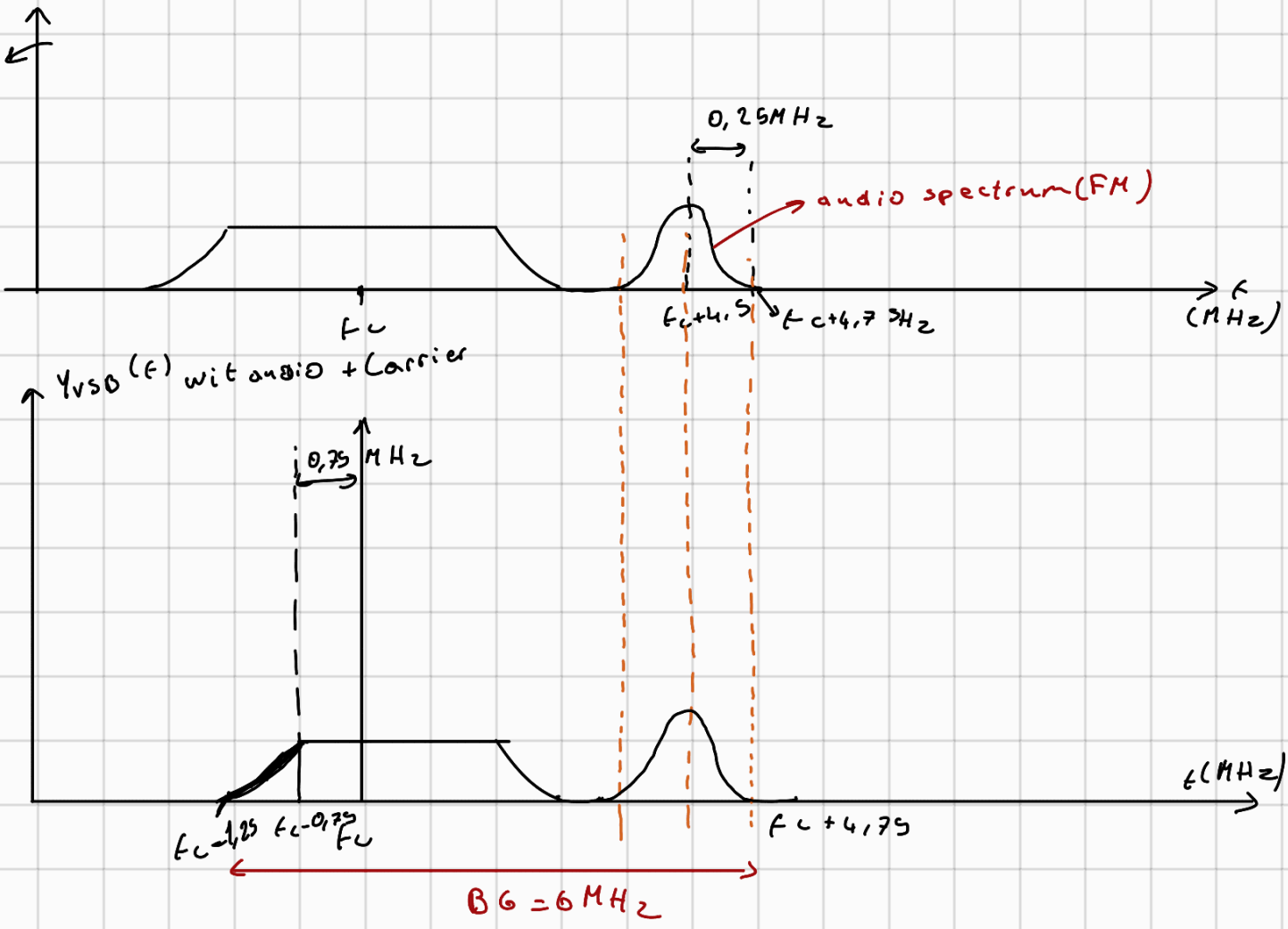
$V_{DSB-LC} > V_{VSB+C} > V_{SSB+C}$

- Temel bant video işareti: (sinyal bant genişliği) = 4,5 MHz

  A4B → BG = 6MHz

Television signal spectra

X(f) with audio spectrum



Potanlama

Frequency Division Multiplexing (FDM)

(Frekans Bölmeli Çoklama - FBÇ)

